PINO: Person-Interaction Noise Optimization for Long-Duration and Customizable Motion Generation of Arbitrary-Sized Groups

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Abstract

Generating realistic multi-person interaction motions that scale to arbitrary group sizes, support fine-grained control, and avoid physical artifacts remains a significant open challenge. Existing two-person diffusion models produce high-quality interactions but cannot generalize beyond their training configuration, while shared-prompt approaches for larger groups yield simplistic, uniform behaviors with frequent body interpenetration. We present ***PINO*** *(**P**erson-**I**nteraction **N**oise **O**ptimization), a training-free framework that composes arbitrary-sized group interactions by decomposing them into sequential, optimized pairwise generations. PINO leverages a pretrained two-person diffusion model (InterGen [14]) with masked conditioning to anchor each new character to an existing "pivot" person, and optimizes the initial Gaussian noise via differentiable physics-based penalties that enforce overlap avoidance, root position control, movement region constraints, and relative orientation. Per-pair text prompts enable distinct interaction roles within a group. For long-duration motion, PINO extends sequences via inpainting with temporal boundary smoothness constraints. Experiments on the InterHuman dataset demonstrate that PINO eliminates body interpenetration (Overlap* 0.000 *vs.* 0.119 *for InterGen) while maintaining comparable motion quality (FID* 13.163 *vs.* 13.278*), scales to five-person groups with consistently low overlap, and generates longer coherent sequences without retraining.*

1. Introduction

Synthesizing realistic human motions for groups of interacting individuals is a fundamental challenge in computer animation, gaming, social robotics, and virtual reality. While text-driven single-person motion generation has seen remarkable progress [3, 5, 22, 27], extending these advances to multi-person settings introduces several compounding difficulties: interactions between individuals must remain spatially coherent; bodies must not overlap or interpenetrate; the complexity of coordination scales with group size; and different roles within an interaction may require different textual descriptions.

Recent two-person interaction models such as InterGen [14] demonstrate impressive quality for dyadic scenarios by training dedicated dual-denoiser networks with mutual attention mechanisms. However, these approaches are architecturally constrained to a fixed number of participants and require paired multi-person training data that is scarce beyond two people. Methods that extend to arbitrary group sizes, such as InterControl [23] and FreeMotion [4], often rely on joint-level control signals or adapt single-person priors, producing interactions that lack the rich mutual dynamics of dedicated interaction models. Supervised multi-person approaches [20] require costly data collection and retraining for each new configuration. A shared text prompt for all participants further encourages uniform, semantically conflated behaviors that fail to capture distinct interaction roles.

An alternative family of methods optimizes the latent noise of pretrained diffusion models at inference time to satisfy user-defined constraints [10, 11, 15, 24]. This paradigm, which we term *noise optimization*, has shown great promise for single-person controllable generation without retraining. Yet it has not been applied to the multi-person interaction setting where physical contact avoidance and inter-person consistency must be jointly enforced.

In this work, we propose **PINO** (**P**erson-**I**nteraction **N**oise **O**ptimization), a training-free framework for generating physically coherent, controllable, and arbitrarily scalable group interactions. Our key insight is that group interactions can be decomposed into semantically coherent pairwise generations anchored by shared pivot characters. By sequentially adding each new person via masked conditioning on a pretrained two-person diffusion model, and optimizing initial noise with differentiable physics penalties, we achieve realistic group interactions without any model retraining. Different text prompts per pair enable nuanced role-specific behaviors within the group. An inpainting-based extension scheme further enables long-duration motion generation with smooth temporal transitions.

Our main contributions are:

- A **training-free framework** that composes arbitrary-sized group interactions from a pretrained two-person dif-

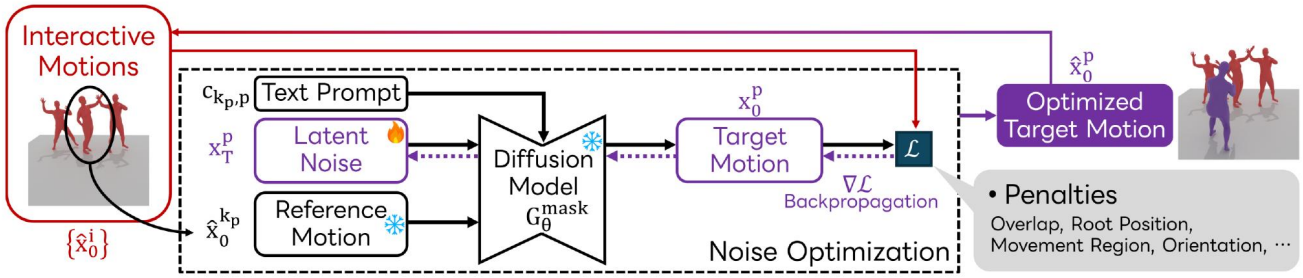

Figure 1. **Person-Interaction Noise Optimization (PINO).** Starting from a two-person generation, PINO incrementally adds each new person by conditioning on the pivot person (person 1) via masked input. The initial noise for each new target is optimized using differentiable physics penalties. Per-pair text prompts allow distinct interaction roles for each newly added person.

fusion model via sequential pairwise generation with per-pair text prompts.

- **Physics-aware noise optimization** that enforces overlap avoidance and fine-grained spatio-temporal controls (root position, movement region, facing orientation, relative positioning) without model retraining.
- **Masked conditioning and pivot-based generation** for clear role assignment, enabling a reference person's motion to guide each new target character's generation within a growing group.
- **Long-duration motion extension** via inpainting with smooth temporal boundary constraints, enabling evolving multi-person interactions over extended time horizons.

Extensive experiments on the InterHuman dataset [14] demonstrate that PINO dramatically reduces body overlap and penetration volume for both two-person and multi-person generation, scales gracefully to five-person groups, and achieves competitive or superior FID scores compared to existing methods.

2. Related Work

Text-to-Motion Generation. Generating 3D human motions from natural language descriptions has progressed rapidly through variational autoencoders [16], flow-based models, and most recently diffusion models [3, 22, 27]. MDM [22] introduced a transformer-based classifier-free diffusion model for human motion, while MLD [3] operates in a compact latent space for efficiency. Methods such as T2M-GPT [26] and MoMask [6] have shown that discrete token-based approaches can rival diffusion quality. For long-duration generation, FlowMDM [1] introduces blended positional encodings to stitch motions smoothly, and T2LM [13] generates long sequences from multi-sentence descriptions. DiffCollage [28] composes large-content generations in parallel through factor graph decompositions, and PriorMDM [19] composes diffusion priors for long sequences and two-person generation. Our work builds on these diffusion advances and extends them to the multi-person regime through noise optimization rather than model redesign.

Multi-Person Interaction Generation. Two-person interaction generation has gained significant attention. InterGen [14] introduces cooperative dual transformers with mutual attention trained on the InterHuman dataset. In2IN [18] augments InterGen with individual-role descriptions to improve intra-person diversity. InterMask [9] applies collaborative masked modeling in discrete space. Tanaka et al. [21] propose role-aware generation separating actor from reactor. For hand interactions, InterHandGen [12] factorizes the joint distribution into conditional single-hand models via conditioning dropout. Extending to more than two persons, ActFormer [25] uses a GAN-Transformer for multi-person action-conditioned generation, and Shan et al. [20] train a concurrent multi-person diffusion model on curated web-scale data. However, supervised approaches require paired multi-person data. FreeMotion [4] unifies single and multi-person generation through conditional distributions but still requires two-person training data for interactions. InterControl [23] takes a zero-shot approach using joint-pair contact constraints with an inverse kinematics module, enabling arbitrary group sizes from single-person priors. PINO differs from all these by leveraging a dedicated two-person prior and composing group motions through sequential noise optimization with physics-based penalties, achieving both high quality and physical plausibility without supervised multi-person training. A detailed comparison is provided in Table 1.

Controllable Motion Generation. Several works address controllable single-person motion generation. GMD [10] and OmniControl [24] incorporate spatial constraints into the diffusion process. ProgMoGen [15] introduces programmable motion generation by optimizing latent codes of a pretrained model against a user-defined error function, enabling open-set motion control for single

Method	#Person in Network Design	#Person in Training Data	Order of Multi-Person Generation	Method
InterControl [23]	One	One	Sequential	ControlNet
FreeMotion [4]	One	Two	Sequential	ControlNet
Shan et al. [20]	Multiple	Multiple	Concurrent	Supervised Training
ProgMoGen [15]	One	Training-free	N/A (Single-Person)	Noise Optimization
Ours	Two (or multiple)	Training-free	Sequential	Noise Optimization

Table 1. Comparison of recent multi-person interaction generation methods, including single-person method ProgMoGen [15].

persons. DNO [11] optimizes diffusion noise for motion editing and control by backpropagating through the full denoising chain. In the image domain, InitNO [7] and Attend-and-Excite [2] optimize initial noise or attention maps for better prompt fidelity. PINO adapts this noise optimization paradigm to the two-person interaction setting, introducing new physics-based penalty terms for group interaction quality.

3. Preliminaries

Denoising Diffusion Probabilistic Models. Diffusion models [8] define a forward process that gradually adds Gaussian noise to data $\mathbf{x}_0$ over T steps: $q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$, where $\{\beta_t\}$ is a fixed noise schedule. The reverse process learns a denoiser $\epsilon_\theta(\mathbf{x}_t, t, c)$ conditioned on context c (e.g., text) to recover $\mathbf{x}_0$ from $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. During inference, starting from $\mathbf{x}_T$, the model iteratively denoises to produce the final sample. The denoiser is trained to predict the noise ϵ added at each step, with the loss $\mathcal{L} = \mathbb{E}_{t,\mathbf{x}_0,\epsilon}[\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t, c)\|^2]$.

InterGen. InterGen [14] is a transformer-based diffusion model for two-person interaction generation. It represents each person's motion as a sequence of 262-dimensional feature vectors encoding joint positions, velocities, rotations, and foot contacts. Two weight-sharing transformer denoisers are coupled via a mutual attention mechanism, allowing each person's denoising process to attend to the other's intermediate states. Text conditioning is provided via a CLIP [17] encoder followed by an additional transformer encoder layer. InterGen is trained on the InterHuman dataset [14], which contains approximately 107M frames of diverse two-person interactions with $23,337$ natural language descriptions.

Motion Representation. Following the HumanML3D representation [5], each person's motion at frame t is represented as a $D = 262$-dimensional vector $\mathbf{m}_t^{(i)}$, where $i \in \{1, 2\}$. This includes the 3D joint positions (22×3), joint velocities, joint rotations, and binary foot contact indicators. The full motion for a pair is concatenated as $\mathbf{m}_t = [\mathbf{m}_t^{(1)}, \mathbf{m}_t^{(2)}] \in \mathbb{R}^{2D}$, yielding a motion tensor of shape $(T, 2D)$ for a sequence of length T.

4. Method

4.1. Overview

PINO generates multi-person interaction motions by incrementally composing pairwise generations using a pretrained two-person diffusion model. Given N persons to generate and per-pair text prompts $\{p_{1,k}\}_{k=2}^{N}$, PINO starts by generating a two-person motion for the initial pair $(1, 2)$. It then conditions on person 1 as the fixed "pivot" to generate person 3, then person 4, and so on, until all N persons are placed. At each step, the initial Gaussian noise for the new target person is optimized to satisfy physics-based constraints. The overall pipeline is illustrated in Fig. 1.

4.2. Masked Conditioning for Pairwise Generation

To add the k-th person ($k \geq 3$) to an existing group, we reformulate the two-person diffusion generation as a conditional generation task. Given the already-generated motion of the pivot person $\hat{\mathbf{m}}^{(1)} \in \mathbb{R}^{T\times D}$, we construct an input to the diffusion model where the reference channel is fixed to the pivot's motion and the target channel is allowed to be denoised freely.

Concretely, we initialize the full two-person noise tensor as:

$$\mathbf{z} = \boldsymbol{\epsilon} \odot (1 - \mathbf{M}) + \hat{\mathbf{m}}^{(1)}_{\text{norm}} \odot \mathbf{M}, \tag{1}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is standard Gaussian noise, $\mathbf{M}$ is a binary mask that selects the feature indices corresponding to the reference (pivot) person's motion, and $\hat{\mathbf{m}}^{(1)}_{\text{norm}}$ is the normalized pivot motion. The reference motion is thus fixed throughout the denoising process, while the target person's motion is denoised conditioned on the pivot via the mutual attention mechanism of InterGen. This formulation requires no modifications to the pretrained model and no additional training.

4.3. Noise Optimization

Directly sampling from the conditioned diffusion model often produces satisfactory motion quality but may violate physical constraints, especially as group size grows and persons are added far from the pivot. We therefore optimize the initial noise $\boldsymbol{\epsilon}$ before the denoising procedure via gradient-based optimization:

$$\boldsymbol{\epsilon}^* = \arg\min_{\boldsymbol{\epsilon}} \mathcal{L}_{\text{total}}(f_\theta(\boldsymbol{\epsilon}; c)), \tag{2}$$

where $f_\theta(\boldsymbol{\epsilon}; c)$ denotes the full denoising function conditioned on text c, and $\mathcal{L}_{\text{total}}$ is a composite of differentiable penalty terms. We use the Adam optimizer to update $\boldsymbol{\epsilon}$ over a fixed number of iterations, backpropagating through the

entire denoising chain. The optimized noise $\boldsymbol{\epsilon}^*$ is then used to generate the final motion via a single forward pass.

4.4. Physics-Based Penalty Terms

Let $\mathbf{J}^{(i)} \in \mathbb{R}^{T\times 22\times 3}$ denote the 3D joint positions extracted from the generated motion of person i, and let $\mathbf{r}^{(i)} \in \mathbb{R}^{T\times 2}$ denote the XZ-plane root (pelvis) trajectory of person i.

Relative Position Penalty $\mathcal{L}_{\text{overlap}}$. To prevent body interpenetration while maintaining plausible proximity, we penalize the root distance between any pair of persons:

$$\mathcal{L}_{\text{overlap}} = \frac{1}{T}\sum_{t=1}^{T}\left[\max(0,\, d_{\min} - \|\mathbf{r}_t^{(1)} - \mathbf{r}_t^{(2)}\|) + \max(0,\, \|\mathbf{r}_t^{(1)} - \mathbf{r}_t^{(2)}\| - d_{\max})\right], \tag{3}$$

where $d_{\min}$ and $d_{\max}$ are minimum and maximum acceptable inter-person distances (default 0.5m and 4.0m). When $k \geq 3$ existing persons are present, additional penalties are applied between the new target and all non-pivot existing persons with threshold 0.5m.

Root Position Penalty $\mathcal{L}_{\text{root}}$. To support user-specified start and end positions for each person, we apply:

$$\mathcal{L}_{\text{root}} = \|\mathbf{r}_0^{(i)} - \hat{\mathbf{r}}_{\text{start}}^{(i)}\|^2 + \|\mathbf{r}_T^{(i)} - \hat{\mathbf{r}}_{\text{end}}^{(i)}\|^2, \tag{4}$$

where $\hat{\mathbf{r}}_{\text{start}}^{(i)}$ and $\hat{\mathbf{r}}_{\text{end}}^{(i)}$ are the target start and end positions in the XZ plane.

Movement Region Penalty $\mathcal{L}_{\text{region}}$. To confine a person's motion within a specified spatial region (circle or axis-aligned rectangle), we penalize the root trajectory leaving the valid region:

$$\mathcal{L}_{\text{region}}^{\text{circle}} = \frac{1}{T}\sum_{t=1}^{T}\max(0,\, \|\mathbf{r}_t^{(i)} - \mathbf{c}\| - R), \tag{5}$$

$$\mathcal{L}_{\text{region}}^{\text{rect}} = \frac{1}{T}\sum_{t=1}^{T}\max(0,\, |r_{t,x}^{(i)} - c_x| - W/2) + \max(0,\, |r_{t,z}^{(i)} - c_z| - H/2), \tag{6}$$

where $\mathbf{c}$ is the region center, R is the radius, and W, H are the rectangle dimensions.

Orientation Penalty $\mathcal{L}_{\text{orient}}$. To control the facing direction of each person, we use the cosine distance between the person's forward direction $\mathbf{n}^{(i)}$ (derived from the hip joints) and a target direction $\hat{\mathbf{n}}^{(i)}$:

$$\mathcal{L}_{\text{orient}} = \frac{1}{T}\sum_{t=1}^{T}\max\left(0,\, \delta - \frac{\mathbf{n}_t^{(i)}\cdot\hat{\mathbf{n}}^{(i)}}{\|\mathbf{n}_t^{(i)}\|\|\hat{\mathbf{n}}^{(i)}\|}\right), \tag{7}$$

Algorithm 1 PINO Multi-Person Generation

Require: Prompts $\{p_{1,k}\}_{k=2}^{N}$, diffusion model f_θ
Generate initial pair: $(\hat{\mathbf{m}}^{(1)}, \hat{\mathbf{m}}^{(2)}) \leftarrow f_\theta(\boldsymbol{\epsilon}^*; p_{1,2})$
Initialize $\mathcal{G} \leftarrow [\hat{\mathbf{m}}^{(1)}, \hat{\mathbf{m}}^{(2)}]$
for $k = 3$ to N **do**
  Initialize noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  Apply masked conditioning with $\hat{\mathbf{m}}^{(1)}$ using Eq. (1)
  Optimize: $\boldsymbol{\epsilon}^* = \arg\min_{\boldsymbol{\epsilon}} \mathcal{L}_{\text{total}}(f_\theta(\boldsymbol{\epsilon}; p_{1,k}), \mathcal{G})$
  Generate: $(_, \hat{\mathbf{m}}^{(k)}) \leftarrow f_\theta(\boldsymbol{\epsilon}^*; p_{1,k})$
  $\mathcal{G} \leftarrow \mathcal{G} \cup \{\hat{\mathbf{m}}^{(k)}\}$
end for
return $\mathcal{G}$

where $\delta = 0.9$ is a cosine similarity threshold.

The total loss is a weighted sum of active penalties:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{overlap}} + \lambda_{\text{root}}\mathcal{L}_{\text{root}} + \lambda_{\text{region}}\mathcal{L}_{\text{region}} + \lambda_{\text{orient}}\mathcal{L}_{\text{orient}}, \tag{8}$$

where penalty terms are enabled selectively based on the user's control requirements.

4.5. Multi-Person Group Generation

Algorithm 1 presents the complete PINO group generation procedure. We maintain a growing motion buffer $\mathcal{G}$ initialized with the pivot person's motion. At each step k, we condition the two-person model on person 1 (the pivot) via masked input (Eq. (1)) and optimize noise (Eq. (2)) with the active penalties. The newly generated target person's motion is appended to $\mathcal{G}$. Per-pair prompts $p_{1,k}$ allow distinct interaction semantics for each pair, *e.g.,* person 2 may "shake hands with person 1" while person 3 "bows toward person 1".

4.6. Long-Duration Motion Extension

For two-person long-duration generation, PINO extends an existing motion sequence $\hat{\mathbf{m}}_{1:T}$ to duration $T' > T$ via iterative inpainting. We generate each new window of length W by conditioning on the last L frames of the existing sequence as a temporal boundary constraint. The initial noise for the new window is optimized with the overlap penalty to prevent interpenetration at the boundary and throughout the extension. Gaussian smoothing ($\sigma = 1$) is applied to the final joint trajectories to ensure smooth transitions between windows.

5. Experiments

5.1. Experimental Setup

Dataset. We conduct all experiments on the **InterHuman** dataset [14], which contains approximately 107M frames of two-person interaction motions with $23,337$ natural language descriptions. We follow the standard train/val/test splits.

Baselines. We compare against (1) **InterGen** [14]: direct two-person generation without noise optimization; (2) **InterGen (inpainting)**: InterGen extended to N persons by treating the first person as a fixed condition and naively inpainting additional persons without noise optimization; and (3) **FreeMotion** [4]: a unified framework supporting number-free text-to-motion synthesis. Our method is denoted **PINO-InterGen**.

Metrics. We evaluate on the following metrics: **FID** (Fréchet Inception Distance) for motion quality; **R-Precision Top-3** for text-motion alignment; **Diversity** for motion variety; **Overlap** (fraction of frames where person roots are closer than 15cm) for physical plausibility; **Penetration Volume (PenVol)** in cm^3 for body interpenetration severity; **Foot Skate** for foot contact realism; and **Max Acceleration (Max Acc.)** for motion smoothness.

Implementation Details. We implement PINO in PyTorch using the publicly available pretrained InterGen checkpoint. Noise optimization runs for $K = 5$ Adam iterations with learning rate $\eta = 0.1$, applied before the main DDIM sampling pass (50 steps, $T = 1000$ diffusion steps). Loss weights are fixed at $\lambda_{\text{root}} = 100$, $\lambda_{\text{region}} = 50$, and $\lambda_{\text{orient}} = 10$; the overlap penalty has unit weight and is always active. For long-duration extension, each new window spans $W = 196$ frames (≈6.5s at 30fps) with a $L = 32$-frame overlap used for boundary inpainting. Gaussian smoothing ($\sigma = 1$) is applied to root trajectories at window junctions to remove discontinuities. All experiments are run on a single NVIDIA A100 GPU; optimizing noise for one additional person in a five-person group takes approximately two minutes.

5.2. Two-Person Interaction with Overlap Avoidance

We first evaluate the two-person case where PINO optimizes the initial noise to enforce the relative position penalty $\mathcal{L}_{\text{overlap}}$. Results are shown in Table 2.

PINO-InterGen reduces Overlap from 0.119 to 0.000 and PenVol from 3112.72 to 275.65 cm^3 (compared to GT 471.75 cm^3), demonstrating that noise optimization effectively eliminates interpenetration. Crucially, semantic quality is preserved: FID improves from 13.278 to 13.163, Diversity improves from 7.793 to 7.904 (vs. GT 7.921), and R-Precision is maintained at 0.675 vs. 0.674. Foot Skate and Max Acceleration remain comparable to the baseline.

5.3. Multi-Person Interaction Generation

Aggregate multi-person comparison. Table 3 compares PINO-InterGen against InterGen (inpainting) for generating a group of persons using person 1 as pivot.

Method	Overlap ↓	PenVol. (cm^3) ↓	Foot Skate ↓	Max Acc. ↓
GT	0.029	471.75	0.099	0.074
InterGen	0.119	3112.72	**0.124**	**0.034**
PINO-InterGen	**0.000**	**275.65**	0.125	**0.034**

(a) Non-semantic metrics.

Method	FID ↓	Div. →	R-Prec. ↑ (Top3)
GT	0.983	7.921	0.715
InterGen	13.278	7.793	0.674
PINO-InterGen	**13.163**	**7.904**	**0.675**

(b) Semantic metrics.

Table 2. Evaluation of two-person interaction generation while avoiding overlap.

Method	Overlap↓	PenVol.↓	Foot Skate↓	Max Acc.↓
InterGen (inpainting)	0.766	25215.78	0.099	**0.030**
PINO-InterGen	**0.000**	**785.01**	**0.091**	0.031

Table 3. Evaluation of multi-person interaction generation

Method	Pair	FID↓	Overlap↓	Foot Skate↓	Max Acc.↓
FreeMotion	(1,2)	19.383	0.255	**0.112**	**0.034**
	(1,3)	25.184	0.806	**0.122**	0.038
	(1,4)	25.812	0.964	0.127	0.040
	(1,5)	25.671	0.991	0.130	0.042
InterGen	(1,2)	13.100	0.119	0.126	**0.034**
	(1,3)	19.414	0.799	0.125	**0.036**
	(1,4)	19.386	0.937	**0.124**	**0.038**
	(1,5)	19.501	0.977	**0.124**	**0.039**
PINO-InterGen	(1,2)	**12.920**	**0.000**	0.127	**0.034**
	(1,3)	**17.170**	**0.002**	0.135	0.040
	(1,4)	**16.890**	**0.019**	0.138	0.043
	(1,5)	**16.911**	**0.069**	0.142	0.045

Table 4. Evaluation of multi-person interaction generation. From pair (1,2), interactions are generated incrementally using 1 as a pivot. Overlap is among all individuals generated up to that step.

PINO-InterGen achieves Overlap of 0.000 vs. 0.766 and PenVol of 785.01 vs. 25215.78 cm^3, representing a 32× reduction in penetration volume. Foot Skate also improves from 0.099 to 0.091.

Scaling to five persons. Table 4 shows pairwise evaluation metrics as we incrementally add persons from pair $(1, 2)$ to $(1, 5)$.

PINO-InterGen maintains near-zero Overlap for pairs $(1, 2)$ through $(1, 4)$ and achieves Overlap 0.069 for pair $(1, 5)$, compared to 0.977 for InterGen and 0.991 for FreeMotion. FID improves consistently: at pair $(1, 5)$, PINO achieves 16.911 vs. 19.501 (InterGen) and 25.671 (FreeMotion). These results confirm that PINO's sequential pivot-based generation composes coherent group interactions at scale. Qualitative comparisons are shown in Fig. 2.

5.4. Long-Duration Motion Extension

We evaluate PINO's ability to extend two-person interactions beyond the training duration. Results are shown in

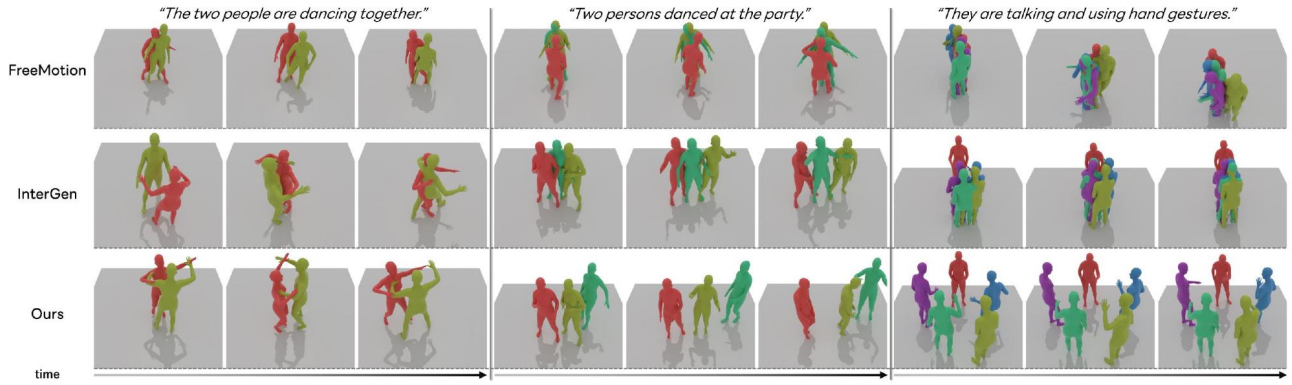

Figure 2. Comparison of generated multi-person interactions. PINO-InterGen (bottom) produces physically plausible motions without body interpenetration, while InterGen (middle) exhibits frequent overlaps. FreeMotion (top) also shows significant overlap artifacts.

Method	Overlap↓	PenVol.↓	Foot Skate↓	Max Acc.↓
InterGen (inpainting)	0.672	15622.16	0.070	0.044
PINO-InterGen	**0.031**	**1375.24**	**0.045**	**0.043**

Table 5. Evaluation of non-semantic elements in motion extension.

Method	FID↓	Overlap↓	Foot Skate↓	Max Acc.↓
InterGen	20.390	0.089	0.081	0.036
PINO-InterGen	**18.930**	**0.0**	**0.054**	**0.025**

Table 6. Evaluation of semantic elements in two-person motion extension. Extended motions are cropped to match the ground truth length for evaluation.

Table 5 (non-semantic) and Table 6 (semantic).

For non-semantic quality, PINO-InterGen reduces Overlap from 0.672 to 0.031, PenVol from 15622.16 to 1375.24 cm^3, and Foot Skate from 0.070 to 0.045. For semantic quality, FID improves from 20.390 to 18.930 and Overlap is eliminated (0.000 vs. 0.089). Max Acceleration also improves from 0.036 to 0.025, indicating smoother transitions between inpainted segments. These results demonstrate that PINO's inpainting extension maintains both physical plausibility and semantic coherence over long sequences.

5.5. Ablation Study

Table 7 ablates the contribution of each penalty term in PINO. We incrementally add penalties and measure positional error (Pos. Err.), Overlap, region violation (Reg. Viol.), and orientation error (Orient. Err.). Positional errors are measured with a 20 cm threshold, region violations with a 10 cm threshold, and orientation errors with a 20-degree threshold.

The baseline InterGen fails all constraints: Overlap 0.292, Reg. Viol. 0.500, Orient. Err. 1.0. Adding $\mathcal{L}_{\text{root}}$ reduces positional error to 0.0 but does not address overlap. Adding $\mathcal{L}_{\text{overlap}}$ eliminates overlap entirely. Adding $\mathcal{L}_{\text{region}}$ reduces region violations from 0.958 to 0.043. Finally, adding $\mathcal{L}_{\text{orient}}$ reduces orientation error from 1.0 to 0.208, confirming each penalty's targeted effect. The qualitative ablation in Fig. 3 further visualizes how each penalty shapes the generated motion.

Method	Pos. Err.↓	Overlap↓	Reg. Viol.↓	Orient. Err.↓
InterGen	1.0	0.292	0.500	1.0
+ $\mathcal{L}_{\text{root}}$	**0.0**	0.333	0.917	1.0
+ $\mathcal{L}_{\text{overlap}}$	**0.0**	**0.0**	0.958	1.0
+ $\mathcal{L}_{\text{region}}$	0.083	**0.0**	**0.043**	1.0
+ $\mathcal{L}_{\text{orientation}}$	0.083	0.043	0.083	**0.208**

Table 7. Ablation study on motion penalties. Metrics include positional errors (Pos. Err.↓), overlap (Overlap↓), region violations (Reg. Viol.↓), and orientation errors (Orient. Err.↓). Positional errors are measured with a threshold of 20 cm, region violations with a threshold of 10 cm, and orientation errors with a threshold of 20 degrees. Lower values indicate better performance.

5.6. Discussion

The sequential pairwise strategy introduces a natural asymmetry in our framework: persons added later must satisfy overlap constraints with respect to all previously generated persons, making each successive addition increasingly constrained. Despite this, FID improves consistently as more persons are added (e.g., 16.911 at pair $(1, 5)$ vs. 13.163 at pair $(1, 2)$ for PINO, compared to a steeper degradation for baselines), suggesting that penalty-guided noise distributions encode stronger interaction priors than naive sampling. We attribute this to the denoiser's learned motion manifold being well-aligned with physically plausible configurations—optimizing toward overlap avoidance effectively steers sampling toward high-quality regions of the learned distribution rather than away from them.

We further observe that the orientation penalty $\mathcal{L}_{\text{orient}}$ achieves the smallest individual improvement among penalty terms (orientation error from 1.0 to 0.208), indicating that facing-direction control is partially decoupled from

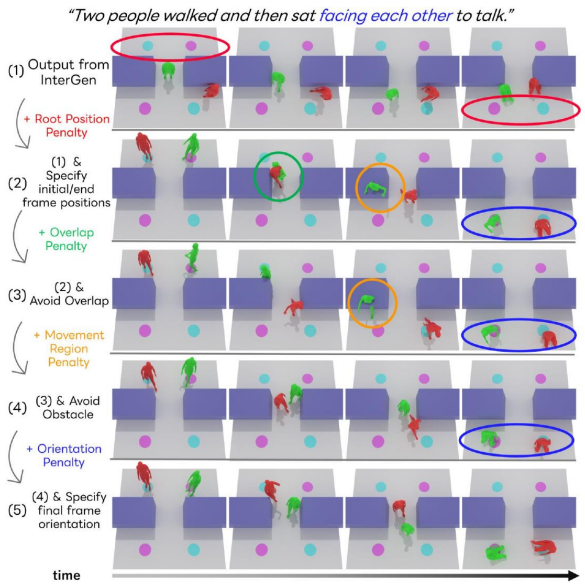

Figure 3. Qualitative ablation of PINO's motion penalties. Each row shows the effect of incrementally adding $\mathcal{L}_{\text{root}}$, $\mathcal{L}_{\text{overlap}}$, $\mathcal{L}_{\text{region}}$, and $\mathcal{L}_{\text{orient}}$.

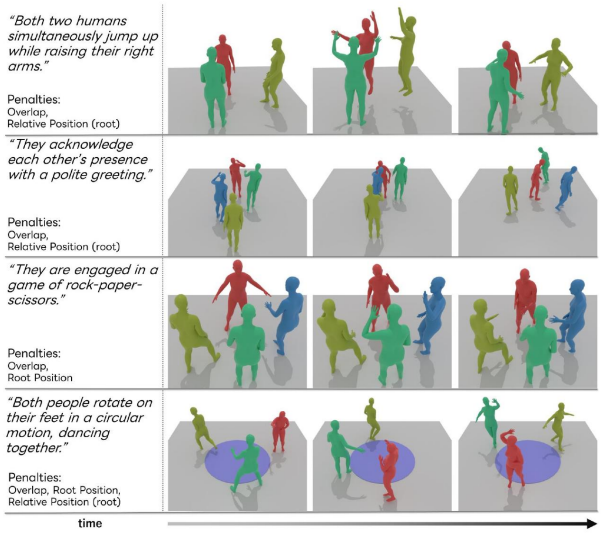

Figure 4. Qualitative results of PINO for multi-person interaction generation with per-pair prompts. Person 1 (pivot) anchors all interactions, while persons 2–5 each exhibit distinct textually-specified behaviors with near-zero physical overlap.

the pivot-conditioned interaction dynamics. This points to a promising research direction: incorporating attention-level orientation guidance rather than purely output-level penalties, which may yield tighter directional control while preserving natural variation in generated motions.

A practical limitation of PINO's sequential generation strategy is that inference time scales linearly with group size: each additional person requires a separate noise optimization pass. For a five-person group, this amounts to approximately eight minutes on a single A100 GPU, which may be prohibitive for real-time applications but remains acceptable for offline animation and cinematic pipelines. Furthermore, since PINO inherits InterGen's fixed-length generation capability, very long sequences require multiple inpainting windows; accumulated smoothing over many windows can gradually attenuate motion expressiveness in later segments. We also observe that PINO performs best when per-pair prompts describe dyadic interactions between two participants. When prompts instead describe highly individual behaviors (*e.g.*, solo gestures unrelated to the pivot), the mutual attention mechanism in InterGen underutilizes the pivot conditioning, leading to weaker spatial coherence. These observations motivate future work on lightweight optimization schedules, parallel multi-person noise optimization to replace the sequential pivot strategy, and alternative two-person backbones with richer cross-person attention that can more strongly leverage pivot motion as an anchor.

5.7. Qualitative Results

Fig. 4 shows qualitative results of PINO for multi-person groups generated with per-pair prompts. Each interaction pair exhibits distinct semantics while physical coherence is maintained across the group.

6. Conclusion

We have presented PINO, a training-free framework for generating physically coherent, customizable, and arbitrarily scalable multi-person interaction motions. By decomposing group interactions into sequential pairwise generations via masked conditioning on a pretrained two-person diffusion model, and optimizing initial Gaussian noise with differentiable physics-based penalties, PINO achieves near-zero body overlap, supports fine-grained spatio-temporal control through per-pair prompts, and extends to long-duration sequences via inpainting. Extensive experiments on the InterHuman dataset demonstrate strong improvements in both physical plausibility and motion quality over existing baselines: overlap is eliminated for two-person generation (0.000 vs. 0.119), penetration volume is reduced by over 90%, and FID improves to 16.911 at five persons vs. 19.501 for InterGen. While orientation error control remains partially effective and inference cost scales with group size due to sequential generation, PINO's modular, training-free design establishes a practical paradigm for compositional multi-person motion generation that can leverage any future two-person diffusion model as a drop-in backbone. Future work could explore simultaneous

multi-person noise optimization to replace the sequential pivot strategy, incorporate richer contact-aware penalties beyond root-level distances, and integrate PINO with large language models for automatic per-pair prompt generation from a single scene description.

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